

Name		Date of Performance	
Roll No.		Date of Submission	
Branch		Examined	

Experiment No:

Title: Linear Measurement by Vernier Calliper, Digital Vernier Calliper and Micrometer

Apparatus:

Instrument	Specifications
Vernier Calliper	L.C. 0.02 mm
Digital Vernier Calliper	L.C. 0.01 mm
Micrometer	L.C.0.01 mm

Introduction:

Linear measuring instruments are designed either for linear measurement (i.e. measuring distance between two lines) or end measurement (i.e. measuring distance between two surfaces).

Line graduated instruments have graduations engraved on the parts of the instruments and distance between these graduations is known. They are used for direct measurement. Measuring accuracy depends on original accuracy of graduations, low efficiency, the instrument is read, definition of graduation lines and general design and workmanship of the device.

Mainly two types:

- 1) Non-precision instruments
- 2) Precision instruments

If the error in the dimension is more than 0.25 mm the instrument producing it are called Non-precision instruments. Precision instruments produce error as less as 0.0025 mm or less in measured dimensions.

A precision measuring instrument should have:

- 1) High degree of sensitivity and accuracy.
- 2) Minimum inertia of moving parts of mechanisms.
- 3) It should eliminate the possibility of error in transferring accurate measurements.
- 4) It should be self-checking and should have good precision.
- 5) Abbes principle should be followed in the design.

VERNIER CALLIPER:

The principle of vernier is that when two scales or division slightly different in size are used, the different between them can be utilized to enhance the accuracy of measurement.

DISCRIPTION: -

The vernier calliper essentially consists of two steel rules and these can slide along each other. The main scale is engraved on a solid L-shaped frame. On this scale graduations are divided into 20 parts so that one small division equals 0.05 mm. One end of the frame contains a fixed jaw, which is shaped into a contact tip as its extremity. The sliding jaw at its left extremity contains another measuring tip. When two measuring tip surfaces are in contact with each other, scale shows zero reading. The adjusting screw can do the finer adjustment of the movable jaw. The locking nuts are adjusted before taking the final reading. The

measuring tips are so designed as to measure inside as well as outside dimensions.

LEAST COUNT: -

The difference between one division on main scale and vernier scale represents the accuracy to which the reading may be taken. This is known as least count of vernier.

$$\text{Least Count (L.C.)} = \frac{\text{Smallest division of main scale}}{\text{Total no. Of divisions on vernier scale}}$$

If the smallest division on main scale is 1 mm and there are 50 divisions on vernier scale then,

$$\text{L.C.} = 1/50 = 0.02 \text{ mm}$$

Total reading on vernier calliper = (Reading on main scale) +
(L.C. $\times$ mark no. of division on vernier scale
that coincides with any mark on main scale)

MICROMETER:

Principle: It consists of employment of a screw and nut, both having accurately cut threads. One complete revolution of the screw will advance it, in relation to the nut threads, a distance equal to lead of the thread.

Construction: Five principle parts 1) Frame 2) Anvil 3) Spindle 4) Sleeve
5) Thimble.

- 1) Frame: The frame is U shaped, at the left end of which an anvil is attached and to the right end, a barrel or sleeve is attached.
- 2) Anvil and spindle: Fixed anvil is at left hand and is screwed in frame. Measuring faces are of hardened steel. The anvil should be ground and lapped flat and parallel to face of spindle.
- 3) Thimble and barrel: The micrometer screw rotates in a fixed nut, which is press, fitted into barrel, which is integral part of frame. The screw on rotations moves in an axial direction relative to the nut. To this screw, spindle is directly attached and the opposite end screw is rigidly attached to thimble. So when thimble rotates and moves axially along the barrel when screw is rotated.
- 4) With main nut, another short nut is attached to it having springs between them to avoid backlash in screw and nut and these two nuts are held by serration like clutch. The shorter nut is rotated and displaced w.r.t. main nut by adjustment for wear of threads.

$$\text{Least Count (L.C.)} = \frac{\text{Smallest division of main scale}}{\text{Total no. of divisions on circular scale}}$$

Reading on micrometer = Reading uncovered on barrel + (L.C. $\times$ No. of divisions on thimble scale which coincides with horizontal line on barrel)

For vernier calliper:

$$\begin{aligned}\text{L.C.} &= \text{Smallest division on main scale} / \text{No. of division on vernier scale} \\ &= 0.5 / 25 = 0.02 \text{ mm}\end{aligned}$$

For micrometer:

$$\text{L.C.} = \text{Pitch of spindle screw} / \text{No. of division on thimble}$$

$$= 0.5 / 50 = 0.01 \text{ mm}$$

OBSERVATION TABLE:

A) For Vernier Calliper

Sr. No.	Dimension	READINGS			AVERAGE
		1	2	3	
1					
2					
3					
4					
5					

B) For Digital Vernier Calliper:

Sr. No.	Dimension	READINGS			AVERAGE
		1	2	3	
1					
2					
3					
4					
5					

C) For micrometer:

Sr. No.	Dimension	READINGS			AVERAGE
		1	2	3	
1					
2					
3					
4					
5					

Result Table:

Sr. No	Vernier Calliper	Digital Vernier Calliper	Micrometer
1.			
2.			
3.			
4.			
5.			
6.			

CONCLUSION:

Vernier calliper and micrometer as precision instruments are studied.

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Experiment No.

Title: Study and use of Sine Bar for Angular Measurement.

AIM-To study the use of sine bar for angular measurement.

Apparatus:

Instrument

Specifications

Industrial Protractor

L.C. 1°

Bevel Protector

L.C. 5'

Sine bar

100 mm & 200 mm roller span

Slip gauges

0.5 to 100 mm, 83 pieces

Dial gauge

L.C. 0.01 mm

Spirit level

0° , 45° , 90°

Surface plate

63 cm x 63 cm

THEORY

UNIVERSAL BEVEL PROTRACTOR-

It is the simplest instrument for measuring angle between two faces of a part such as taper shaft, bushing, etc. It has two arms which can be set along the faces and a circular scale to indicate the angle between them. Accuracy of the instrument is increased by adding vernier. The body of the instrument is extended from one of the arms called stock.

Other arm is in the form of a blade. The working edge of stock is about 150 to 300 mm long and 13 mm wide and 2 mm thick. Its ends are leveled at 45 to 60 within an accuracy of 5 minutes of an hour.

The blade rotates on a turret mounted on the body and can be damped in any position by a locking nut. Either the turret or body carries the divided circle (main circle) and other member carries a vernier.

· **LEAST COUNT-** 12 divisions on either side of the vernier scale coincide with 23 divisions on either side of main scale.

One division on either side of vernier scale = $23 / 12$ ie: one division on vernier is 5' short = 1 degree 55' than 2' on main scale. Least count = 5'

Total angle reading = main scale reading + LC (considering vernier division)

Or, least count = smallest division on main scale / total divisions on vernier scale

$$= 1 \text{ degree} / 12 \text{ divisions} = 60 / 12 = 5'$$

WORKING

For measuring the included angle in big jobs such as V-blocks as shown in fig. The bevel protractors are placed on the flat surface of the job and the long adjustable block is rotated in a direction with turret which gives the increase of cycle angle on main scale and finally the long adjustable blade is adjusted parallel to the surface whose angle with the horizontal is to be measured.

Then the main scale reading and coincident vernier scale readings are taken and angle is measured. For measuring angle on a small workpiece has a special attachment called as 'acute angle attachment' is provided with bevel protractor. Workpieces, which are very small, are directly placed on the acute angle measurement and direct adjusting the long blade does measurement.

SINE BAR

A sine bar is a carefully machined tool steel bar with two perfectly scaled cylinders that that may or may not be fastened to it. Its mandatory that –

1. The rollers are of the same diameter.
2. Their centre distance is absolutely correct.
3. Centre line of rollers is parallel to edge of bar.

PRINCIPLE

It forms hypotenuse of the right-angled triangle and slip gauge (or measuring height) forms the side of opposite to angle being measured.

Thus, $\sin x = h / l = h / 100$

$$x = \sin^{-1} (h / 100)$$

STEPS TO FOLLOW WHILE USING SINE BAR

1. FOR LIGHT WORKPIECE – Initially the approximate angle with the help of bevel protractor is found. Then a sine bar is set at this angle and clamped to the angle plate. Next workpiece is clamped to sine bar. Now dial indicator is mounted on a stand. Its carried in a surface gauge on surface plate and moved along the sine bar. Adjusting the slip gauge reflects variations if any. However if there is variation the corresponding error in angle is obtained by noting the variation over a measured distance of travel and converting it to angular measurement. Thus if a pointer moves 0.05 mm over a distance of 80 mm.

2. FOR HEAVY WORKPIECE- The work piece is placed on a surface plate. Sine bar is kept on it. Now instead of the work piece sine bar work datum surface is made perfectly parallel to the surface plate by method explained below-

In the fig, the sine bar is mounted on heavy components. The height over rollers can be measured by vernier height gauge. A fiducial indicator dial gauge can be used to ensure constant measurement.

The anvil of height gauge is adjusted with probe of dial test gauge so that the dial gauge shows the sum of readings for top work position of roller.

$$x = \sin^{-1} (h/l)$$

LOCATING THE JOB AT A GIVEN ANGLE

Here instead of finding x , we find h .

$h = l \sin x$. A slip gauge block of height h is inspected above the roller and the job mounted on the inclined sine bar.

<u>OBSERVATION TABLE</u>					
Sr. NO	JOB NO.	MEASUREMENT			ERROR
		INDUSTRIAL PROTECTOR	BEVEL PROTRACTOR	BY SINE BAR	

SAMPLE CALCULATION

CONCLUSION

The sine bar is more accurate than Industrial protractor and bevel protractor because the sine bar can be located very accurately with a slip gauge block. The unknown values are found by trigonometry. Sine tables are highly accurate for small jobs.

Questions:

1. Using a 100 mm sine bar and a 200 mm sine bar. Will it make any deviation in the results? If so, give the reasons.
2. Will the roller contact point distance of a sine bar vary when the sine bar is mounted at different inclination?
3. Explain why it is not preferred to use Sine Bar for measuring angle more than 45° .

Questions:

1. What is the difference between precision and non-precision instruments?
2. Write the procedure of calculating Least Count of Vernier calliper and micrometer.
3. What is the principle of Vernier calliper / Micrometer?
4. State the errors during measurement by Vernier calliper.
5. What is depth micrometer?